Part 2: Quantitative data analysis

Quantitative Session of the 2026 HUSOE ELPS Methods and Data Institute

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0.1 Case Study: HBCUs and Educational Geography

In this section, we analyze data on Historically Black Colleges and Universities (HBCUs) to explore patterns in location, institutional type, and founding history.

0.1.1 Step 1: Load Packages

```r
#install.packages("tidyverse")
#install.packages("lubridate")
library(tidyverse)
library(lubridate)
```

We'll start with a guided example using data on Historically Black Colleges and Universities.

0.1.2 Step 2: Load the data

```r
library(tidyverse)
hbcu <- read_csv("https://raw.githubusercontent.com/quant-shop/intro-comp-educ-soc/refs/heads/r

# View the first few rows
head(hbcu)
```

```
# A tibble: 6 x 7
  name                        city       state founded   lat   lon type
  <chr>                       <chr>      <chr>   <dbl> <dbl> <dbl> <chr>
1 Alabama A&M University      Normal     AL       1875  34.8 -86.6 Public, 4 Year
2 Alabama State University    Montgomery AL       1867  32.4 -86.3 Public, 4 Year
3 Albany State University     Albany     GA       1903  31.6 -84.2 Public, 4 Year
4 Alcorn State University     Lorman     MS       1871  31.9 -91.1 Public, 4 Year
5 Allen University            Columbia   SC       1870  34.0 -81.0 Private, 4 Year
6 American Baptist College    Nashville  TN       1924  36.2 -86.8 Private, 4 Year
```

```r
# Check structure
glimpse(hbcu)
```

```
Rows: 102
Columns: 7
$ name    <chr> "Alabama A&M University", "Alabama State University", "Albany ~
$ city    <chr> "Normal", "Montgomery", "Albany", "Lorman", "Columbia", "Nashv~
$ state   <chr> "AL", "AL", "GA", "MS", "SC", "TN", "AR", "SC", "NC", "FL", "A~
$ founded <dbl> 1875, 1867, 1903, 1871, 1870, 1924, 1884, 1870, 1873, 1904, 19~
$ lat     <dbl> 34.7834, 32.3643, 31.5785, 31.8769, 34.0298, 36.1659, 34.7465,~
$ lon     <dbl> -86.5683, -86.2952, -84.1543, -91.1458, -81.0115, -86.7844, -9~
$ type    <chr> "Public, 4 Year", "Public, 4 Year", "Public, 4 Year", "Public,~
```

```r
# Summary statistics
summary(hbcu)
```

```
     name               city              state              founded
 Length:102         Length:102         Length:102         Min.   :1837
 Class :character   Class :character   Class :character   1st Qu.:1870
 Mode  :character   Mode  :character   Mode  :character   Median :1886
                                                          Mean   :1895
                                                          3rd Qu.:1905
                                                          Max.   :1988
      lat             lon              type
 Min.   :18.34   Min.   :-98.50   Length:102
 1st Qu.:32.48   1st Qu.:-90.13   Class :character
```

```
 Median :34.02   Median :-84.64   Mode  :character  
 Mean   :34.31   Mean   :-85.13                     
 3rd Qu.:36.17   3rd Qu.:-80.78                     
 Max.   :39.93   Max.   :-64.96                     
```

0.1.3 Step 3: Clean the data

```r
hbcu <- hbcu %>%
  mutate(
    founded = as.numeric(founded),
    type = as.factor(type),
    state = as.factor(state)
  )
```

0.1.4 Step 4: Basic Exploration

How many institutions are in the data set?

```r
nrow(hbcu)
```

```
[1] 102
```

What is the distribution by HBCU type?

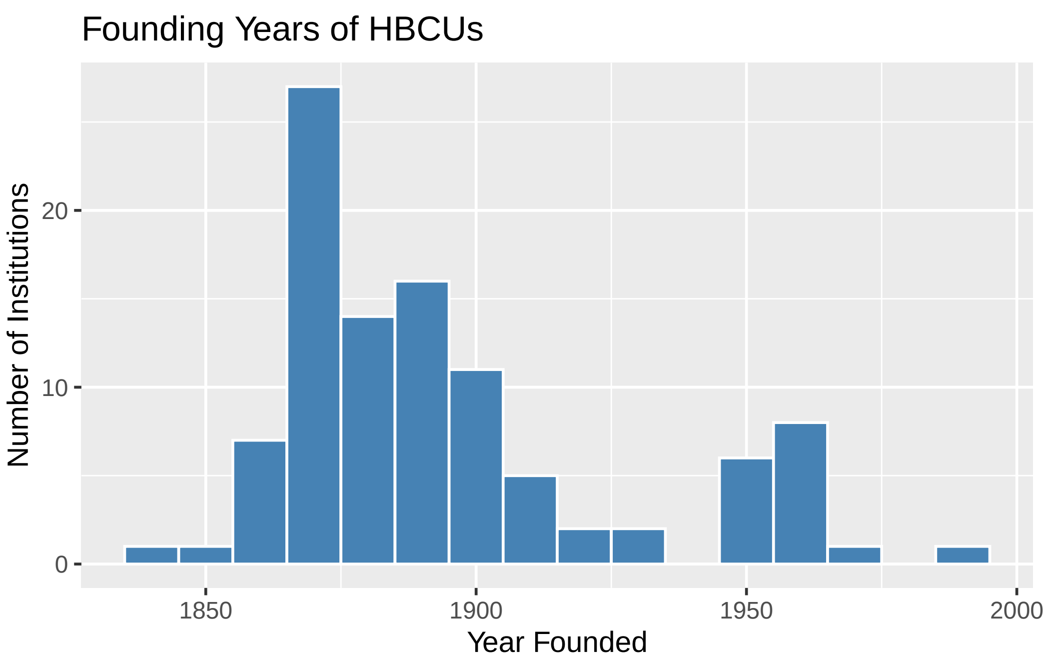
```r
hbcu %>%
  count(type) %>%
  arrange(desc(n))
```

```
# A tibble: 5 x 2
  type                    n
  <fct>               <int>
1 Private, 4 Year        45
2 Public, 4 Year         40
3 Public, 2 Year         11
4 Private, Specialized    4
5 Private, 2 Year         2
```

0.1.5 Step 5: Historical Analysis

When were the HBCUs founded?

```r
hbcu %>%
  ggplot(aes(x = founded)) +
  geom_histogram(binwidth = 10, fill = "steelblue", color = "white") +
  labs(
    title = "Founding Years of HBCUs",
    x = "Year Founded",
    y = "Number of Institutions"
  )
```



0.1.6 Step 6: Geographic Distribution

HBCUs by state

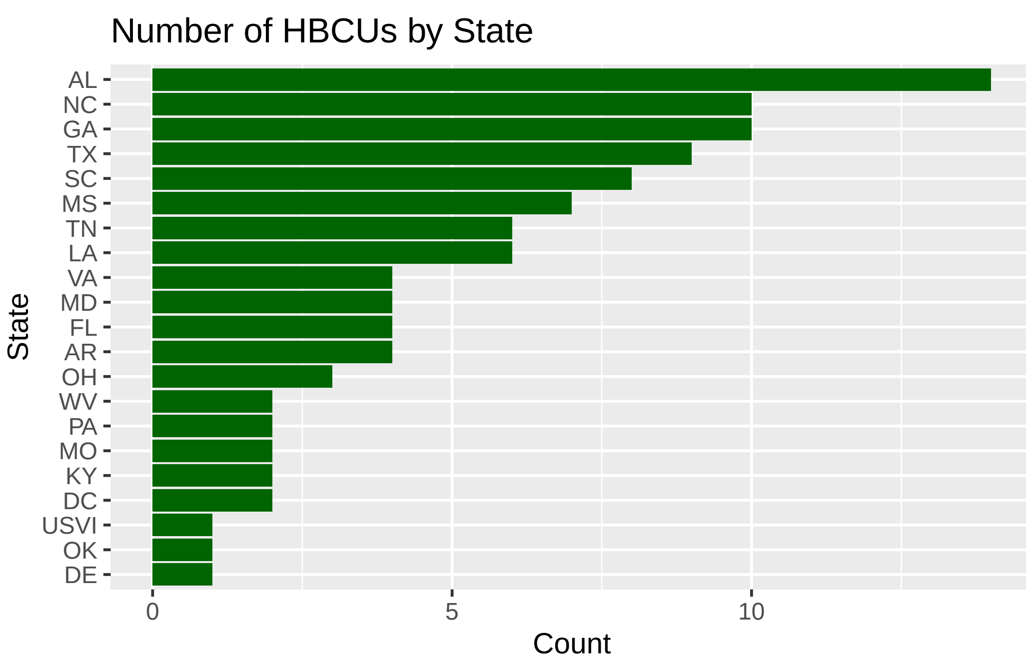
```r
hbcu %>%
  count(state, sort = TRUE)
```

```
# A tibble: 21 x 2
   state     n
   <fct> <int>
 1 AL       14
 2 GA       10
 3 NC       10
 4 TX        9
 5 SC        8
 6 MS        7
 7 LA        6
 8 TN        6
```

```
 9 AR          4
10 FL          4
# i 11 more rows
```

Basic visualization of HBCUs by state

```
hbcu %>%
  count(state) %>%
  ggplot(aes(x = reorder(state, n), y = n)) +
  geom_col(fill = "darkgreen") +
  coord_flip() +
  labs(
    title = "Number of HBCUs by State",
    x = "State",
    y = "Count"
  )
```

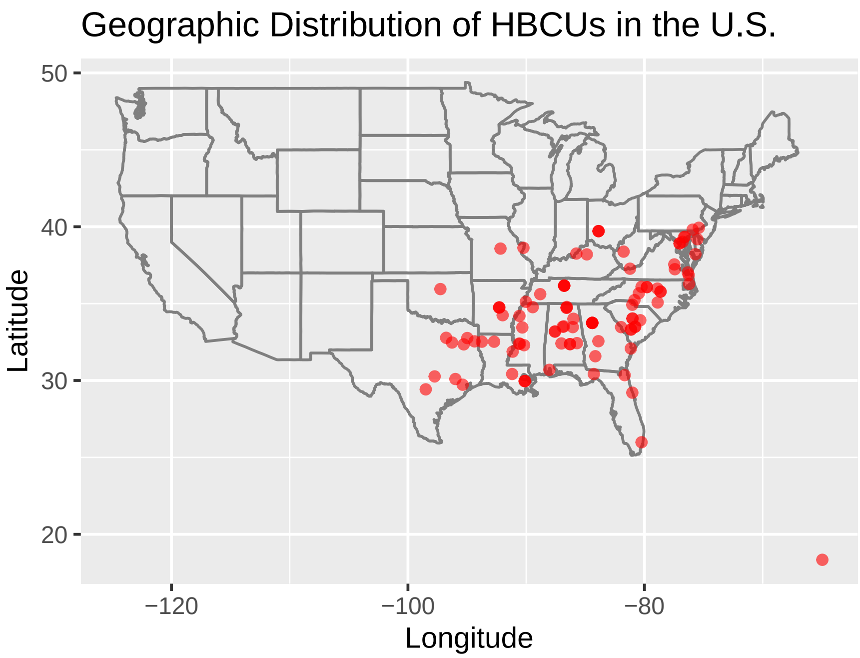


0.1.7 Step 7: Mapping

```
#install.packages("maps", repos = "http://cran.us.r-project.org")
library(maps)
```

```
hbcu %>%
  ggplot(aes(x = lon, y = lat)) +
  borders("state") +
  geom_point(color = "red", alpha = 0.6) +
```

```
  coord_fixed(1.3) +
  labs(
    title = "Geographic Distribution of HBCUs in the U.S.",
    x = "Longitude",
    y = "Latitude"
  )
```



0.2 Case Study: Data on State School Districts

0.2.1 Step 1: Load the libraries

```
library(dplyr)
library(readr)
library(stringr)
library(janitor)
```

0.2.2 Step 2: Load the data

```
district <- read.csv("https://raw.githubusercontent.com/quant-shop/intro-comp-educ-soc/refs/hea
# district <- read.csv("../data/state-district-data.csv")

str(district)
```

```
'data.frame':	57 obs. of  36 variables:
 $ State.Name                                                      : chr  "ALABAMA" "ALASKA"
 $ State.Abbr                                                      : chr  "AL " "AK " "AS "
 $ ANSI.FIPS.State.Code..State..Latest.available.year              : int  1 2 60 4 5 59 6 8
 $ Total.Number.Operational.School.Districts..District..2023.24    : int  156 54 1 685 304 
 $ Total.Number.Operational.Schools..Public.School..2023.24        : int  1524 494 29 2427 
 $ Total.Number.Operational.Charter.Schools..Public.School..2023.24 : chr  "18" "31" "†" "57
 $ Prekindergarten.Students..State..2023.24                        : chr  "21924" "3320" "-
" "20810" ...
 $ Kindergarten.Students..State..2023.24                           : chr  "55881" "9257" "-
" "72320" ...
 $ Grades.1.8.Students..State..2023.24                             : chr  "447317" "78890" 
" "654843" ...
 $ Grades.9.12.Students..State..2023.24                            : chr  "223528" "39776" 
" "369124" ...
 $ Grade.13.Students..State..2023.24                               : chr  "†" "†" "†" "†" .
 $ Ungraded.Students..State..2023.24                               : chr  "†" "†" "†" "533"
 $ American.Indian.Alaska.Native.Students..State..2023.24          : chr  "5997" "28284" "-
" "46878" ...
 $ Asian.or.Asian.Pacific.Islander.Students..State..2023.24        : chr  "11016" "6458" "-
" "35975" ...
 $ Hispanic.Students..State..2023.24                               : chr  "84196" "10076" "
" "539087" ...
 $ Black.or.African.American.Students..State..2023.24              : chr  "236304" "2955" "
" "64938" ...
 $ White.Students..State..2023.24                                  : chr  "381550" "61502" 
" "378320" ...
 $ Nat..Hawaiian.or.Other.Pacific.Isl..Students..State..2023.24    : chr  "831" "4148" "-
" "4017" ...
 $ Two.or.More.Races.Students..State..2023.24                      : chr  "28756" "17820" "
" "48415" ...
 $ Total.Enrollment..Exclude.AE..for.SY.2014.15.onward..State..2023.24: chr  "748650" "131243"
" "1117630" ...
 $ Full.Time.Equivalent..FTE..Teachers..State..2023.24             : chr  "42857.92" "7221.
" "49260.47" ...
 $ Pupil.Teacher.Ratio..State..2023.24                             : chr  "17.47" "18.17" "
" "22.69" ...
 $ State.Name..State..2023.24                                      : chr  "ALABAMA" "ALASKA"
 $ Total.Number.of.Districts.with.Enrollment..District..2023.24    : int  150 54 0 664 260 
 $ Total.Number.of.School.Districts..District..2023.24             : int  159 54 1 722 308 
 $ Total.Number.of.Public.Schools..Public.School..2023.24          : int  1527 498 29 2557 
 $ Male.Students..State..2023.24                                   : chr  "384500" "67646" 
 $ Female.Students..State..2023.24                                 : chr  "364150" "63597" 
 $ Black.or.African.American...male..State..2023.24                : chr  "120605" "1488" "
" "33191" ...
 $ Black.or.African.American...female..State..2023.24              : chr  "115699" "1467" "
" "31747" ...
 $ White...male..State..2023.24                                    : chr  "197384" "31786" 
```

```
" "194292" ...
 $ White...female..State..2023.24                                  : chr  "184166" "29716" "
" "184028" ...
 $ Grade.12.Students...Black.or.African.American..State..2023.24   : chr  "15413" "241" "-
" "5876" ...
 $ Secondary.Teachers..State..2023.24                              : chr  "20188.71" "3673.4
" "15996.72" ...
 $ Elementary.Teachers..State..2023.24                             : chr  "18395.46" "2985.0
" "30478.09" ...
 $ Full.Time.Equivalent..FTE..Staff..State..2023.24                : chr  "79963.90" "16423
" "109199.43" ...
```

0.2.3 Step 3: Clean the data

```
district_clean <- district %>%

  # Fix missing/suppressed values
  mutate(across(where(is.character), ~na_if(., "-"))) %>%
  mutate(across(where(is.character), ~na_if(., "†"))) %>%
  mutate(across(where(is.character), str_trim)) %>%

  # Convert numeric columns safely
  mutate(across(
    where(is.character),
    ~parse_number(.)
  ))
```

0.2.4 Step 4: Feature engineering

We can create variables that answer educational questions.

```
str(district_clean)
```

```
'data.frame':	57 obs. of  36 variables:
 $ State.Name                                                      : num  NA NA NA NA NA NA
  ..- attr(*, "problems")= tibble [57 x 4] (S3: tbl_df/tbl/data.frame)
  .. ..$ row     : int [1:57] 1 2 3 4 5 6 7 8 9 10 ...
  .. ..$ col     : int [1:57] NA NA NA NA NA NA NA NA NA NA ...
  .. ..$ expected: chr [1:57] "a number" "a number" "a number" "a number" ...
  .. ..$ actual  : chr [1:57] "ALABAMA" "ALASKA" "AMERICAN SAMOA" "ARIZONA" ...
 $ State.Abbr                                                      : num  NA NA NA NA NA NA
  ..- attr(*, "problems")= tibble [57 x 4] (S3: tbl_df/tbl/data.frame)
  .. ..$ row     : int [1:57] 1 2 3 4 5 6 7 8 9 10 ...
  .. ..$ col     : int [1:57] NA NA NA NA NA NA NA NA NA NA ...
  .. ..$ expected: chr [1:57] "a number" "a number" "a number" "a number" ...
```

```
 .. ..$ actual  : chr [1:57] "AL" "AK" "AS" "AZ" ...
$ ANSI.FIPS.State.Code..State..Latest.available.year                : int  1 2 60 4 5 59 6 8
$ Total.Number.Operational.School.Districts..District..2023.24       : int  156 54 1 685 304 
$ Total.Number.Operational.Schools..Public.School..2023.24           : int  1524 494 29 2427 
$ Total.Number.Operational.Charter.Schools..Public.School..2023.24   : num  18 31 NA 571 105 
$ Prekindergarten.Students..State..2023.24                          : num  21924 3320 NA 208
$ Kindergarten.Students..State..2023.24                             : num  55881 9257 NA 723
$ Grades.1.8.Students..State..2023.24                               : num  447317 78890 NA 6
$ Grades.9.12.Students..State..2023.24                              : num  223528 39776 NA 3
$ Grade.13.Students..State..2023.24                                 : num  NA NA NA NA NA NA 
$ Ungraded.Students..State..2023.24                                 : num  NA NA NA 533 174 
$ American.Indian.Alaska.Native.Students..State..2023.24            : num  5997 28284 NA 468
$ Asian.or.Asian.Pacific.Islander.Students..State..2023.24          : num  11016 6458 NA 359
$ Hispanic.Students..State..2023.24                                 : num  84196 10076 NA 53
$ Black.or.African.American.Students..State..2023.24                : num  236304 2955 NA 64
$ White.Students..State..2023.24                                    : num  381550 61502 NA 3
$ Nat..Hawaiian.or.Other.Pacific.Isl..Students..State..2023.24       : num  831 4148 NA 4017 
$ Two.or.More.Races.Students..State..2023.24                        : num  28756 17820 NA 48
$ Total.Enrollment..Exclude.AE..for.SY.2014.15.onward..State..2023.24: num  748650 131243 NA 
$ Full.Time.Equivalent..FTE..Teachers..State..2023.24               : num  42858 7221 NA 492
$ Pupil.Teacher.Ratio..State..2023.24                               : num  17.5 18.2 NA 22.7 
$ State.Name..State..2023.24                                        : num  NA NA NA NA NA NA 
 ..- attr(*, "problems")= tibble [57 x 4] (S3: tbl_df/tbl/data.frame)
 .. ..$ row     : int [1:57] 1 2 3 4 5 6 7 8 9 10 ...
 .. ..$ col     : int [1:57] NA NA NA NA NA NA NA NA NA NA ...
 .. ..$ expected: chr [1:57] "a number" "a number" "a number" "a number" ...
 .. ..$ actual  : chr [1:57] "ALABAMA" "ALASKA" "AMERICAN SAMOA" "ARIZONA" ...
$ Total.Number.of.Districts.with.Enrollment..District..2023.24       : int  150 54 0 664 260 
$ Total.Number.of.School.Districts..District..2023.24               : int  159 54 1 722 308 
$ Total.Number.of.Public.Schools..Public.School..2023.24            : int  1527 498 29 2557 
$ Male.Students..State..2023.24                                     : num  384500 67646 NA 5
$ Female.Students..State..2023.24                                   : num  364150 63597 NA 5
$ Black.or.African.American...male..State..2023.24                  : num  120605 1488 NA 33
$ Black.or.African.American...female..State..2023.24                : num  115699 1467 NA 31
$ White...male..State..2023.24                                      : num  197384 31786 NA 1
$ White...female..State..2023.24                                    : num  184166 29716 NA 1
$ Grade.12.Students...Black.or.African.American..State..2023.24      : num  15413 241 NA 5876 
$ Secondary.Teachers..State..2023.24                                : num  20189 3673 NA 159
$ Elementary.Teachers..State..2023.24                               : num  18395 2985 NA 304
$ Full.Time.Equivalent..FTE..Staff..State..2023.24                  : num  79964 16423 NA 10
 ..- attr(*, "problems")= tibble [1 x 4] (S3: tbl_df/tbl/data.frame)
 .. ..$ row     : int 51
 .. ..$ col     : int NA
 .. ..$ expected: chr "a number"
 .. ..$ actual  : chr "‡"
```

```
# what kinds of questions might you answer with this data set?
```

0.3 Recommended readings

Knaflic, C. N. (2015). *Storytelling with data: A data visualization guide for business professionals.* Wiley.